SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration, Data transformation and data Reduction , for effective data preparation (BL3,BL4)

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks: 50**

Annexure B

Analytical Problem Solving

Criteria	*Problem Understanding (2)*	*Method Selection (2)*	*Solution Process (2.5)*	*Accuracy of Calculation (2)*	*Interpretation of Results (1.5)*	*Total(10)*
Weightage	*20%*	20%	25%	20%	15%	***100%***
Q1						
Q2						
Q3						
Q4						
Q5						

Question 1: Covariance Analysis in Health Insurance

The table below shows the annual premium increase rate (xi) and the number of claims per 100 policyholders (yi) in a health insurance company. Using the covariance formula, determine whether premium increase and claim frequency have a positive or inverse relationship. Before computing covariance, calculate the mean of x and y.

Premium Increase % (xi)	Claims per 100 Policyholders (yi)
3.2	15
4	18
5.5	25
4.8	20

Step 1: Calculate the Mean of x and y

$\bar{x} = (3.2 + 4.0 + 5.5 + 4.8) / 4$

$= 17.5 / 4$

$= 4.375$

$\bar{y} = (15 + 18 + 25 + 20) / 4$

$= 78 / 4$

$= 19.5$

Step 2: Compute the Deviations and Their Products

xi	yi	$xi - \bar{x}$	$yi - \bar{y}$	$(xi - \bar{x})(yi - \bar{y})$
3.2	15	-1.175	-4.5	5.2875
4	18	-0.375	-1.5	0.5625
5.5	25	1.125	5.5	6.1875
4.8	20	0.425	0.5	0.2125

Step 3: Sum the Products

$\Sigma(xi - \bar{x})(yi - \bar{y})$

$= 5.2875 + 0.5625 + 6.1875 + 0.2125$

$= 12.25$

Step 4: Apply the Covariance Formula

$Cov(x, y) = \Sigma(xi - \bar{x})(yi - \bar{y}) / n$

$= 12.25 / 4$

$= 3.0625$

(If the sample covariance, dividing by $n - 1 = 3$, is used instead: $Cov(x, y) = 12.25 / 3 = 4.0833$ — either denominator gives a positive result.)

Step 5: Interpretation

Since $Cov(x, y) = 3.0625$ is positive, there is a positive (direct) relationship between the premium increase rate and the number of claims per 100 policyholders — as the premium increase percentage rises, the claim frequency also tends to rise.

Question 2: Data Smoothing using Equal-Frequency Binning (Healthcare) (5 Marks)

A hospital is analyzing patient blood pressure readings (systolic) to identify trends. Due to minor fluctuations, the analyst performs local smoothing using equal-frequency (equi-depth) binning.

Sample Data Set (27 values, already sorted): 110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

Step 1: Decide the Bin Depth

With 27 data values, an equal-frequency (equi-depth) partition of bin depth 3 is used, so that each bin contains exactly 3 consecutive sorted values and the data divides evenly into 9 bins.

Step 2: Partition the Sorted Data into Equal-Frequency Bins

Bin	Values in Bin
Bin 1	110, 112, 115
Bin 2	115, 118, 120
Bin 3	120, 122, 124
Bin 4	124, 126, 126
Bin 5	126, 126, 130
Bin 6	135, 135, 138
Bin 7	138, 138, 138
Bin 8	140, 145, 150
Bin 9	152, 160, 180

Step 3: Smoothing by Bin Means

Each value in a bin is replaced by the mean (average) of that bin:

Bin	Values	Sum	Bin Mean
Bin 1	110, 112, 115	337	112.33
Bin 2	115, 118, 120	353	117.67
Bin 3	120, 122, 124	366	122.00
Bin 4	124, 126, 126	376	125.33
Bin 5	126, 126, 130	382	127.33
Bin 6	135, 135, 138	408	136.00
Bin 7	138, 138, 138	414	138.00
Bin 8	140, 145, 150	435	145.00
Bin 9	152, 160, 180	492	164.00

Step 4: Smoothed (Reduced) Data Set

Replacing every value in each bin with that bin's mean gives the smoothed data set:

112.33, 112.33, 112.33, 117.67, 117.67, 117.67, 122.00, 122.00, 122.00, 125.33, 125.33, 125.33, 127.33, 127.33, 127.33, 136.00, 136.00, 136.00, 138.00, 138.00, 138.00, 145.00, 145.00, 145.00, 164.00, 164.00, 164.00

Question 3: Equal-Width Binning for Risk Categorization (5 Marks)

An insurance company evaluates Body Mass Index (BMI) of 18 clients for risk classification. To simplify analysis, they smooth the data by replacing values with the closest bin boundaries using equal-width partitioning.

Sample Data Set (Sorted BMI, 18 values): 18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

Step 1: Determine the Range

Minimum value = 18.5, Maximum value = 39.5

Range = Max − Min = 39.5 − 18.5 = 21.0

Step 2: Decide the Number of Bins and Compute Bin Width

Using 4 equal-width bins (matching 4 natural risk categories — e.g., Low, Moderate, High, Very High):

Bin Width = Range / Number of Bins

= 21.0 / 4

= 5.25

Step 3: Determine the Bin Boundaries

Bin	Lower Boundary	Upper Boundary
Bin 1	18.50	23.75
Bin 2	23.75	29.00
Bin 3	29.00	34.25
Bin 4	34.25	39.50

Step 4: Assign Each Value to Its Bin and Smooth by Boundaries

Each value is replaced by whichever bin boundary (lower or upper) it is numerically closer to:

BMI Value	Bin Range	Dist. to Lower	Dist. to Upper	Smoothed Value
18.5	[18.5, 23.75]	0.00	5.25	18.5
19.2	[18.5, 23.75]	0.70	4.55	18.5
21.8	[18.5, 23.75]	3.30	1.95	23.75
24.9	[23.75, 29.0]	1.15	4.10	23.75
25.5	[23.75, 29.0]	1.75	3.50	23.75
26.2	[23.75, 29.0]	2.45	2.80	23.75
26.4	[23.75, 29.0]	2.65	2.60	29.0
27.8	[23.75, 29.0]	4.05	1.20	29.0
29.2	[29.0, 34.25]	0.20	5.05	29.0
30.2	[29.0, 34.25]	1.20	4.05	29.0
30.4	[29.0, 34.25]	1.40	3.85	29.0
31.9	[29.0, 34.25]	2.90	2.35	34.25
32.4	[29.0, 34.25]	3.40	1.85	34.25
33.1	[29.0, 34.25]	4.10	1.15	34.25
33.6	[29.0, 34.25]	4.60	0.65	34.25
34.7	[34.25, 39.5]	0.45	4.80	34.25
38.2	[34.25, 39.5]	3.95	1.30	39.5
39.5	[34.25, 39.5]	5.25	0.00	39.5

Step 5: Smoothed (Boundary-Reduced) Data Set

Final smoothed BMI values: 18.5, 18.5, 23.75, 23.75, 23.75, 23.75, 29.0, 29.0, 29.0, 29.0, 29.0, 34.25, 34.25, 34.25, 34.25, 34.25, 39.5, 39.5

Question 4: Data Reduction using Bin Medians (Medical Supply Costs) (5 Marks)

A hospital administration is reviewing the cost of medical consumables (in ₹). To reduce data for mining purposes, they discretize the sorted price records using bin medians.

Sample Data Set (Costs in ₹, 27 values): 120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

Step 1: Sort the Data Set

Before binning, the raw values are arranged in ascending order:

120, 140, 145, 145, 148, 150, 150, 150, 155, 155, 155, 155, 160, 160, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

Step 2: Partition into Equal-Frequency Bins (Depth = 3)

With 27 sorted values, dividing into bins of depth 3 gives 9 equal-frequency bins:

Bin	Values in Bin
Bin 1	120, 140, 145
Bin 2	145, 148, 150
Bin 3	150, 150, 155
Bin 4	155, 155, 155
Bin 5	160, 160, 165
Bin 6	170, 170, 175
Bin 7	175, 175, 175
Bin 8	180, 190, 200
Bin 9	210, 230, 300

Step 3: Find the Median of Each Bin

Since each bin already contains 3 sorted values, the median is simply the middle value:

Bin	Values	Bin Median
Bin 1	120, 140, 145	140
Bin 2	145, 148, 150	148
Bin 3	150, 150, 155	150
Bin 4	155, 155, 155	155
Bin 5	160, 160, 165	160
Bin 6	170, 170, 175	170
Bin 7	175, 175, 175	175
Bin 8	180, 190, 200	190
Bin 9	210, 230, 300	230

Step 4: Reduced (Smoothed) Data Set

Replacing every value in each bin with that bin's median gives the data-reduced set:

140, 140, 140, 148, 148, 148, 150, 150, 150, 155, 155, 155, 160, 160, 160, 170, 170, 170, 175, 175, 175, 190, 190, 190, 230, 230, 230

This reduces the 27 original price records to 9 distinct representative values (one repeated median per bin), achieving data reduction while preserving the overall cost distribution.

Question 5: Standardization of Patient Age Data (5 Marks)

Given the following ages of insured individuals: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28. Standardize the variable by performi ng the following: (a) Compute the mean absolute deviation (MAD) of age. (b) Compute the z-score for the first four observations.

Part (a): Mean Absolute Deviation (MAD)

Step 1: Compute the Mean

$\bar{x} = (18+22+25+42+28+43+33+35+56+28) / 10 = 330 / 10 = 33$

Step 2: Compute Absolute Deviations $|xi - \bar{x}|$

xi	18	22	25	42	28	43	33	35	56	28
$\|xi - \bar{x}\|$	15.0	11.0	8.0	9.0	5.0	10.0	0.0	2.0	23.0	5.0

Step 3: Sum the Absolute Deviations and Divide by n

$\Sigma|xi - \bar{x}| = 15 + 11 + 8 + 9 + 5 + 10 + 0 + 2 + 23 + 5 = 88$

$MAD = \Sigma|xi - \bar{x}| / n = 88 / 10 = 8.8$

Part (b): Z-score for the First Four Observations

Using the mean absolute deviation as the measure of spread (robust z-score), the standardized value is computed as:

$z = (xi - \bar{x}) / MAD$

xi	$xi - \bar{x}$	$z = (xi - \bar{x}) / MAD$
18	$18 - 33 = -15$	-1.7045
22	$22 - 33 = -11$	-1.2500
25	$25 - 33 = -8$	-0.9091
42	$42 - 33 = 9$	1.0227

Interpretation

A negative z-score indicates the age is below the mean age of 33, while a positive z-score indicates it is above the mean, each expressed in units of the mean absolute deviation (8.8 years). For example, the first observation (age 18) lies about 1.70 MAD-units below the mean age, while the fourth observation (age 42) lies about 1.02 MAD-units above the mean age.

Code of Conduct:

*I **Syed Arshad Khadri (192525255)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

RUBRICS FOR CALCULATION:

Numerical Problems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation